PHYSICS **Max Marks: 60**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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18. A body travel uniformly a distance of $(s \pm \Delta s)$ in a time $(t \pm \Delta t)$. what may be the condition so that body within the error limits move with a velocity $\left(\frac{s}{t} \pm \frac{\Delta s}{\Delta t}\right)$?

A) $\frac{\Delta t}{t} + \frac{s(\Delta t)^2}{(\Delta s)t^2} = \pm 1$ **B)** $\frac{\Delta t}{t} + \frac{s(\Delta t)}{(\Delta s)t} = \pm 1$

C) $\frac{\Delta t}{t} + \frac{(\Delta s)t}{(\Delta t)s} = \pm 1$ **D)** $\frac{\Delta t}{t} + \frac{s^2(\Delta t)}{(\Delta s)^2 t} = \pm 1$

19. A current i flows through a wire shaped in the form of an infinite parabola of Latus rectum 4a , then magnetic field at the focus of the parabola is

A) $B = \frac{\mu_0 i}{2a}$ **B)** $B = \frac{\mu_0 i}{8a}$ **C)** $B = \frac{\mu_0 i}{4a}$ **D)** $B = \frac{\mu_0 i}{a}$

20. A uniformly charged ring of mass M_0 and radius R_0 is placed on a smooth horizontal surface, The total charge on the ring is Q_0 . A small point charge of mass m_0 and charge q_0 is placed at the centre of the ring. The particle is displaced by a very small distance of $x(x << R_0)$ from the centre in the plane of the ring and released, The ring is not fixed and free to move on the horizontal surface. Find the time after which particle returns to the centre again

A) $T = 2\pi\sqrt{\frac{M_0 m_0 .2R_0^3}{(M_0+m_0)4\pi \in_0 Q_0 q_0}}$ **B)** $T = \pi\sqrt{\frac{M_0 m_0 .8\pi \in_0 R_0^3}{(M_0+m_0)Q_0 q_0}}$

C) $T = \frac{\pi}{4}\sqrt{\frac{M_0 m_0 .2R_0^3}{(M_0+m_0)\pi \in_0 Q_0 q_0}}$ **D)** $T = 2\pi\sqrt{\frac{M_0 m_0 .2\pi \in_0 R_0^3}{(M_0+m_0)Q_0 q_0}}$

21. A uniform disc with radius r and mass m can roll without slipping on a cylindrical surface and is attached to a bar ABC with length L and negligible mass. The bar is attached at a point A to a spring with a constant K and can rotate freely about point B in the vertical plane. knowing that end A is given a small displacement and released. Then the frequency of the resulting vibration in terms of m,L,k and g is

(18) $V = \frac{S}{t}$

$$\frac{\Delta V}{V} = \frac{\Delta S}{S} + \frac{\Delta t}{t}$$

$$\Delta V = \frac{\Delta S}{S} \times V + \frac{\Delta t}{t} \times V$$

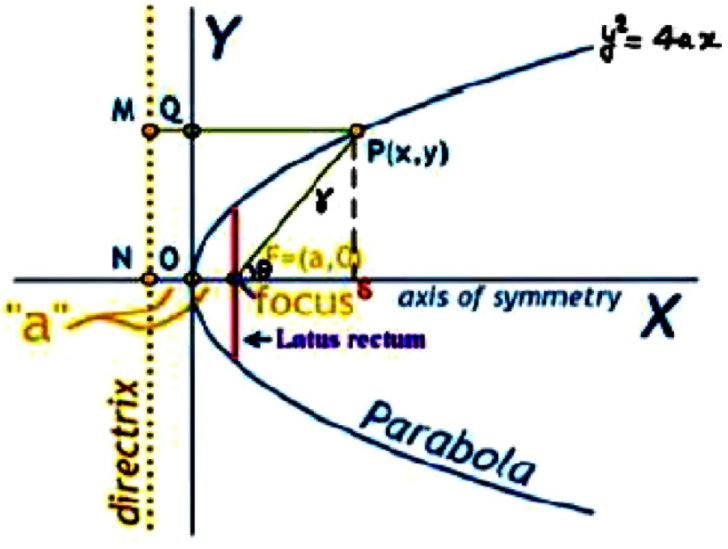
$$= \frac{\Delta S}{t} + \frac{\Delta t\, S}{t^2}$$

Here $\Delta V = \frac{\Delta S}{\Delta t}$

$$\Rightarrow \frac{\Delta S}{\Delta t} = \frac{\Delta S}{t} + \frac{\Delta t\, S}{t^2}$$

$$\Rightarrow 1 = \frac{\Delta t}{t} + \frac{(\Delta t)^2 S}{\Delta S\, t^2}$$

(19) Geometry of Parabola



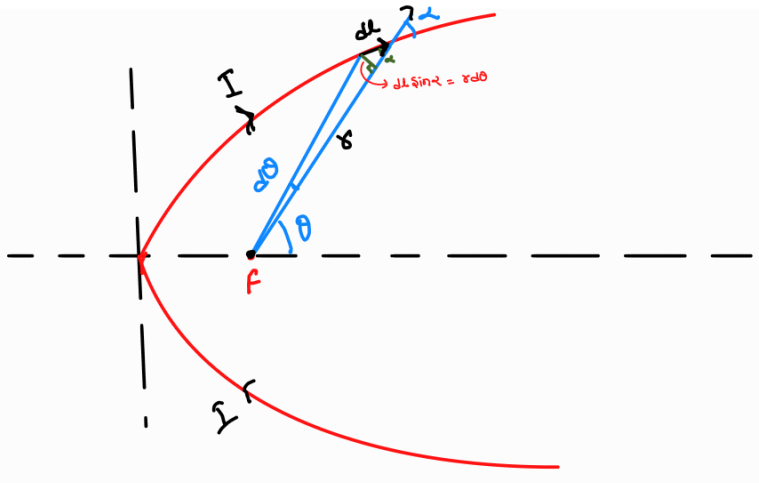
γ = distance between point on parabola and focus.

PF = PM

$\gamma = \gamma\cos\theta + 2a$

$\boxed{\gamma(1-\cos\theta) = 2a}$ $\Rightarrow \gamma = \frac{2a}{1-\cos\theta}$

Parametric eqⁿ of Parabola



$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I\, d\vec{l} \times \vec{\gamma}}{\gamma^3}$$

$$= \frac{\mu_0}{4\pi} \frac{I\, dl\, \gamma \sin\alpha}{\gamma^3 \gamma^2}$$

Using Biot-Savart law

$$dB = \frac{\mu_0 I\, dl \sin\alpha}{4\pi\, \gamma^2}$$

$$= \frac{\mu_0 I\, (\gamma\, d\theta)}{4\pi\, \gamma^2}$$

$$= \frac{\mu_0 I}{4\pi\gamma}\, d\theta$$



$$= \frac{\mu_0 I}{4\pi \times 2a}(1-\cos\theta)\,d\theta$$

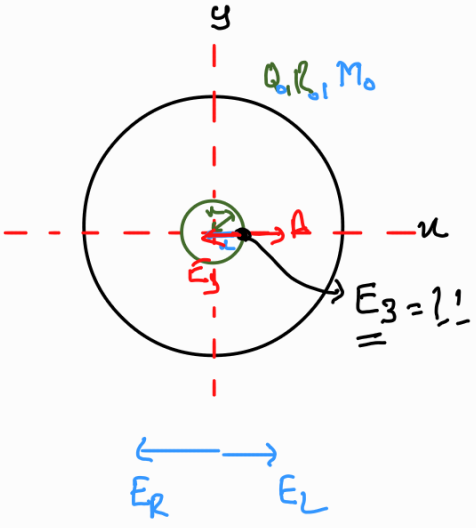
$$B = \frac{\mu_0 I}{8\pi a}\int_0^{2\pi}(1-\cos\theta)\,d\theta$$

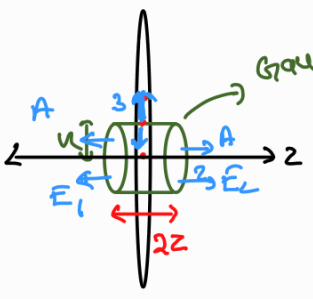
$$= \frac{\mu_0 I}{8\pi a}\left[\theta - \sin\theta\right]_0^{2\pi}$$

$$= \frac{\mu_0 I}{8\pi a}\left[2\pi - 0\right]$$

$$= \frac{\mu_0 I}{4a}$$

(20) Ef at a distance x from centre in plane of ring





z is very small.

x is also very small.

$$|E_1| = |E_2| = \frac{KQ_0 z}{(R_0^2 + z_0^2)^{3/2}} \approx \frac{KQ_0 z}{R_0^3}$$

$E_R > E_L$

$E_3 \to E_R$

Net flux through Gaussian surface

$$\phi = \phi_1 + \phi_2 + \phi_3$$

$$= E_1 \pi x^2 \cos 0^\circ + E_2 \pi x^2 \cos 0^\circ + E_3 2\pi x\, 2z \cos 180^\circ$$

$$= 2 \times \frac{KQ_0 z}{R_0^3}\pi x^2 - E_3 \times 4\pi x z$$

as $q_{enc} = 0$

$$\Rightarrow \phi = 0$$

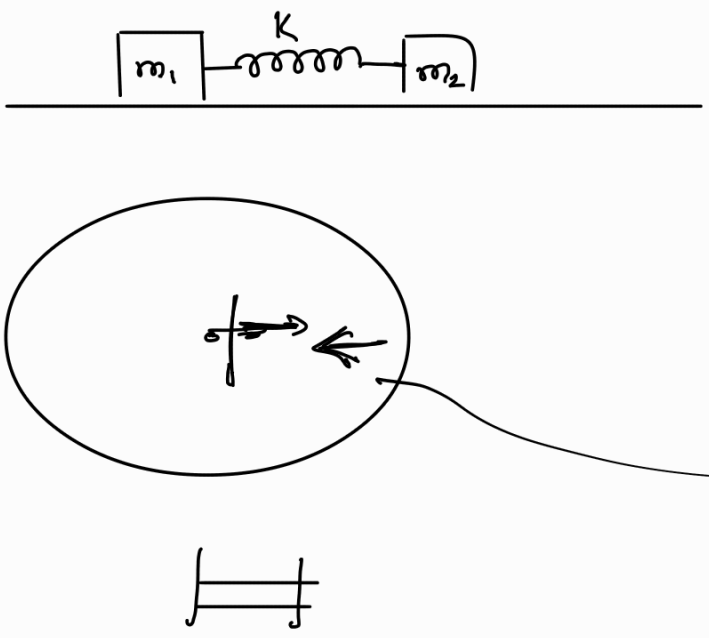
$$\Rightarrow \frac{2KQ_0 z}{R_0^3}\pi x^2 - E_3 \times 4\pi x z = 0$$

$$\Rightarrow E_3 = \frac{KQ_0 x}{2R_0^3}$$

Force on charge is $-q_0E_3$

$$F = -q_0E_3 = -\frac{KQ_0q_0}{2R_0^3}x \qquad \text{(SHM equation)}$$

It will behave as 2 block & spring system



But here

$$K_{eff} = \frac{KQ_0q_0}{2R_0^3}$$

reduced mass,

$$\mu = \frac{M_0m_0}{M_0+m_0}$$

time period

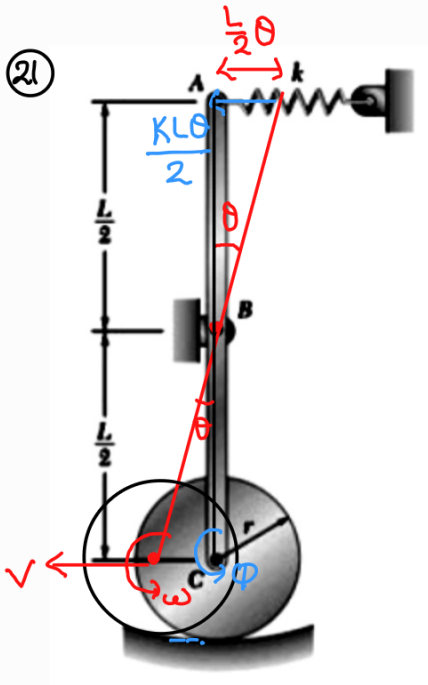
$$T = 2\pi\sqrt{\frac{\mu}{K_{eff}}}$$

$$= 2\pi\sqrt{\frac{m_0M_0}{m_0+M_0}\times\frac{2R_0^3}{KQ_0q_0}}$$

It will return to centre after time,

$$t = \frac{T}{2} = \pi\sqrt{\frac{m_0M_0}{m_0+M_0}\times\frac{2R_0^3}{KQ_0q_0}}$$

(21)



Let Disc is rotated by ϕ

then, $\frac{L}{2}\theta = r\phi$

$$\phi = \frac{L\theta}{2r} \Rightarrow \frac{d\phi}{dt} = \frac{L}{2r}\frac{d\theta}{dt}$$

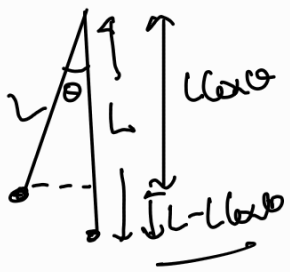
$$\Rightarrow \omega = \frac{L}{2r}\frac{d\theta}{dt}$$

& $V = \omega r$

Energy of System



$$U = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 + \frac{1}{2}kx^2 + mg\frac{L}{2}(1-\cos\theta)$$



$$= \frac{1}{2}m(\omega r)^2 + \frac{1}{2}\frac{mr^2}{2}\omega^2 + \frac{1}{2}K\left(\frac{L\theta}{2}\right)^2 + mg\frac{L}{2}(1-\cos\theta)$$

$$= \frac{m\omega^2r^2}{2} + \frac{m\omega^2r^2}{4} + \frac{K}{8}L^2\theta^2 + mg\frac{L}{2}(1-\cos\theta)$$

$$= \frac{3m\omega^2r^2}{4} + \frac{K}{8}L^2\theta^2 + mg\frac{L}{2}(1-\cos\theta)$$

$$\frac{dU}{dt} = \frac{3mr^2}{4} \times 2\omega \times \frac{d\omega}{dt} + \frac{K}{8}L^2 \times 2\theta \times \frac{d\theta}{dt} + mg\frac{L}{2}\sin\theta\frac{d\theta}{dt}$$

$$0 = \frac{3mr^2\omega}{4} \times \frac{L}{2r}\frac{d^2\theta}{dt^2} + \frac{KL^2}{8}\theta \times \frac{2r\omega}{L} + mg\frac{L}{2}\theta \times \frac{2r\omega}{L}$$

$$\Rightarrow \frac{3mL}{8}\frac{d^2\theta}{dt^2} = -\frac{KL}{4}\theta - mg\theta$$

$$\Rightarrow \frac{d^2\theta}{dt^2} = -\left(\frac{2K}{3m} + \frac{4g}{3L}\right)\theta \qquad \alpha = -\omega^2\theta$$

$$\omega = \sqrt{\frac{2K}{3m} + \frac{4g}{3L}}$$

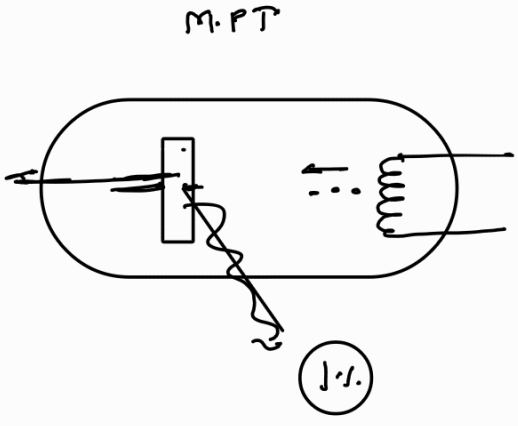
$$f = \frac{\sqrt{\frac{2K}{3m} + \frac{4g}{3L}}}{2\pi}$$

(22) $P = VI = 50 \times 20\,W = 1000\,W$

$P_{target} = 1000\,W \times 99\% = 990\,W = 990\,J/S$

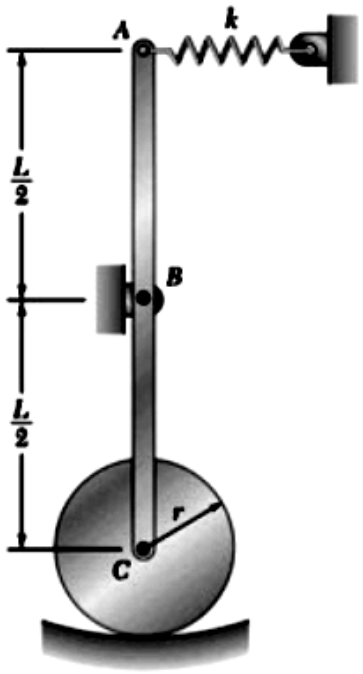
$ms\Delta T = 990$

$$\Delta T = \frac{990}{ms} = \frac{990}{1 \times \frac{425}{5}} {}^{10}\,{}^{\circ}C/sec = 2\,{}^{\circ}C/S$$



$$\lambda_{min} = \frac{hc}{eV} = \frac{12400}{50 \times 10^3}\,A^{\circ} = 0.248\,A^{\circ}$$

Target should have high melting pt, high thermal conductivity.



A) $f = \dfrac{\sqrt{\dfrac{4k}{3m} + \dfrac{2g}{3L}}}{2\pi}$

B) $f = \dfrac{\sqrt{\dfrac{2k}{3m} + \dfrac{4g}{3L}}}{2\pi}$

C) $f = \dfrac{\sqrt{\dfrac{2k}{3m} + \dfrac{4g}{L}}}{2\pi}$

D) $f = \dfrac{\sqrt{\dfrac{2k}{m} + \dfrac{4g}{3L}}}{\pi}$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

22. An x-ray tube is operating at 50 KV and 20 mA. The target material of the tube has a mass of 1 Kg and specific heat of $495 Jkg^{-1}\,{}^{0}C^{-1}$. one percent of the supplied electric power is converted into x-rays and entire remaining energy goes into heating the target. Then

A) The average rate of rise of temperature of the target would be $2^{0}C$ / sec

B) The minimum wavelength of x-ray emitted is about $0.25\ X\ 10^{-10}\ m$

C) A suitable target material must have high melting temperature.

D) A suitable target material must have low thermal conductivity

23. Two loudspeakers A and B are located 20 m apart and emit sound of frequencies 118 Hz and 121Hz respectively. A car moves along the perpendicular bisector PQR of line AB

where Q is the midpoint of AB. Points P and R lie on opposite sides of Q at a distance of 1800 m from Q. The car starts from point P, crosses Q and then reaches R. Let f(t) represent the beat frequency measured by a person sitting in the car. If the car moves with a uniform speed 60 km/h and the speed of sound is 330 m/s, then the correct statements from the following are

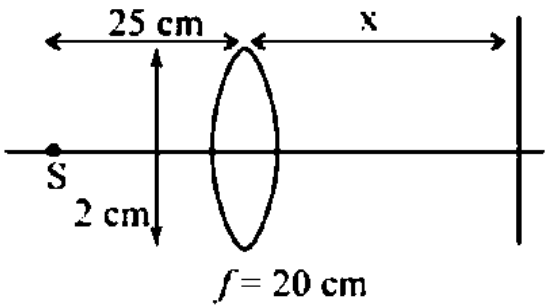
A) Rate of change of f is greatest when the car crosses Q

B) Rate of change of f is least when the car crosses Q

C) $f_p + f_R$ equals four times f_Q

D) $f_p - f_R$ approximately equals one tenth of f_Q

24. A point source of light S is placed on the axis of a thin lens of focal length 20 cm as shown. A screen is placed normal to the axis of lens at a distance X from it. Treat all rays as paraxial.



A) As x is increased from zero , intensity continuously decreases

B) As x is increased from zero ,intensity first increases then decreases

C) Intensity at centre of screen for X=90 cm and X= 110 cm is same

D) Radius of bright circle obtained on screen is equal to 1 cm for X = 200 cm

SECTION-III

(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreenvirtual numeric keypad in the place designated to enter the answer.

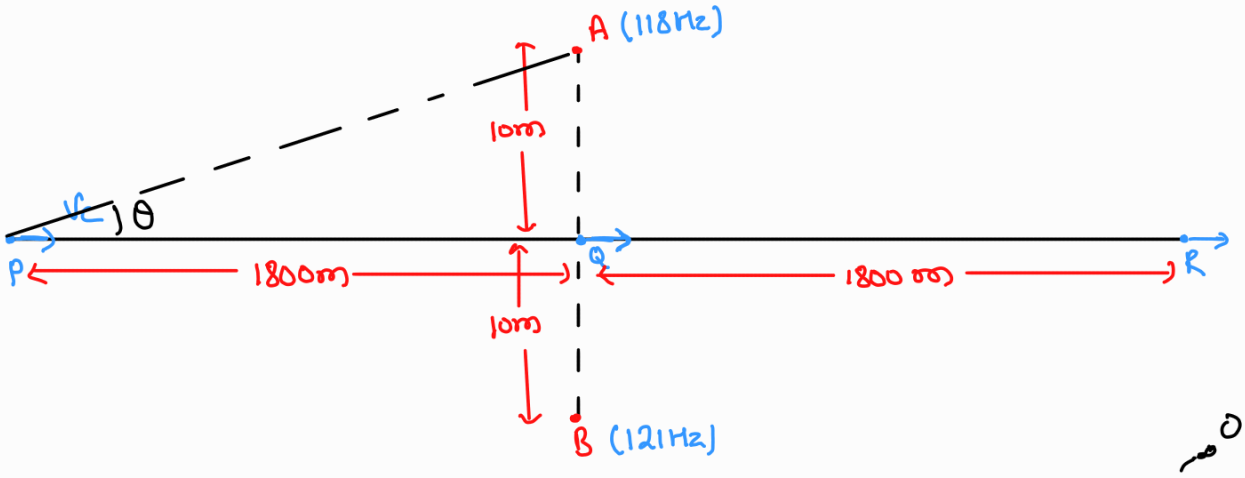
•Answer to each question will be evaluated according to the following marking scheme

Full Marks :+4 If ONLY the correct integer is entered;Zero Marks : 0 In all other cases..

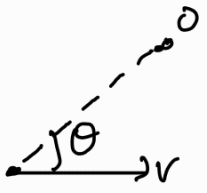
25. A gas consisting of rigid diatomic molecules was expanded in a poly trophic process so that the rate of collisions of the molecules against the vessel's wall did not change. Find the molar heat capacity of the gas in this process is found to be XR. The value of X is_____

26. A Structure consisting of two identical square plates of side $l = 40\ cm$ hinged to a common axle at one of their edges is placed on a fixed horizontal cylinder of radius r =10

(23)



$v_c = 60\,km/h = 60 \times \frac{5}{18} = \frac{50}{3}\,m/s = 16.67\,m/s$



$v_s = 330\,m/s$

$$f_P = \frac{v_s + v_c\cos\theta}{v_s} f_B - \frac{v_s + v_o\cos 2\theta}{v_s} f_A$$

$$= \frac{v_s + v_c\cos\theta}{v_s}(f_B - f_A)$$

$$= \left(1 + \frac{v_c}{v_s}\cos\theta\right)(121-118)$$

$$= \left(1 + \frac{50}{3\times 330} \times 1\right) \times 3$$

$$= \frac{990+50}{3\times 330} \times 3$$

$$= \frac{1040}{330} = \frac{104}{33} = 3.15\,Hz$$

$$f_R = \left(\frac{v_s - v_c\cos\theta}{v_s}\right) \times f_B - f_A$$

$$= \frac{990-50}{3\times 330} \times 3 = \frac{940}{330} = \frac{94}{33} = 2.85\,Hz$$

$f_Q = 121 - 118 = 3\,Hz$

* $f_P + f_R = 6\,Hz = 2f_Q$

* $f_P - f_R = 3.15 - 2.85 = 0.3\,Hz = \frac{f_Q}{10}$

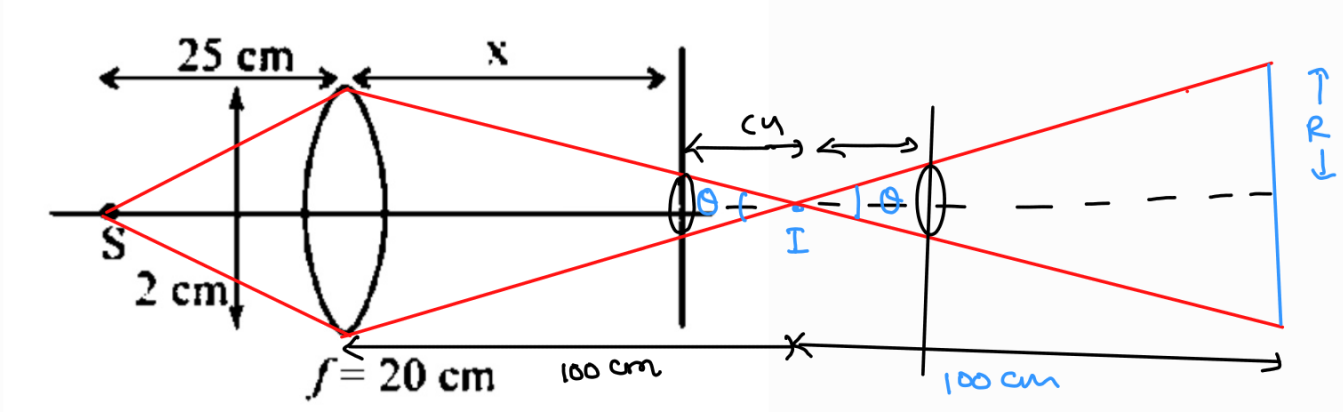
* Q charge fastest at Q $\Rightarrow$ f will change fatest at Q.

(24) $\frac{1}{V} - \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \frac{1}{V} - \frac{1}{-25} = \frac{1}{20}$

$\Rightarrow \frac{1}{V} = \frac{1}{20} - \frac{1}{25} = \frac{5-4}{100} = \frac{1}{100}$

$\Rightarrow V = 100\,cm$



* as screen is moved intensity will 1st increase then decrease

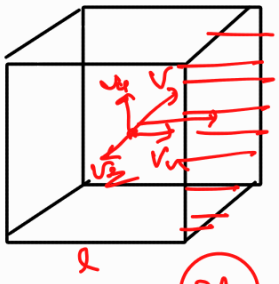
* at x = 90 cm & x = 110 cm Intensity will be same as they are at same distance from I.

* R = 1 cm at x = 200 cm

(25) Rate of Collision with wall $= \frac{1}{6} n v_{rms}$

$n = \frac{N}{V}$

$= \frac{1}{2} \frac{N}{V} \times \sqrt{\frac{3RT}{M}} = \text{Const}$



$\Rightarrow \frac{\sqrt{T}}{V} = \text{Const}$

$\Rightarrow \frac{T}{V^2} = \text{Const}$

$\Rightarrow \frac{\left(\frac{PV}{nR}\right)}{V^2} = \text{Const}$

$\Rightarrow \frac{P}{V} = \text{Const}$

$\Rightarrow PV^{-1} = \text{Const}$

$\Delta t = \frac{2l}{v_x}$

$v_x = v/3$

$\frac{N \times v_{rms}}{3 \times 2l}$

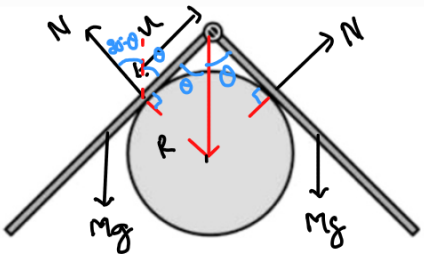
$\frac{1}{6} \frac{N v_{rms}}{l A}$

$\frac{1}{6} \times \left(\frac{N}{V}\right) v_{rms}$

$C = C_V + \frac{R}{1-n}$

$= \frac{5R}{2} + \frac{R}{1-(-1)} = \frac{5R}{2} + \frac{R}{2} = 3R$

(26)



$\tan\theta = \frac{R}{u}$

Balancing the forces

$N \sin\theta = Mg \quad -(i)$

Balancing torques

$Nu = Mg \frac{L}{2} \sin\theta \quad -(ii)$

from (i) & (ii)

$\frac{Mg}{\sin\theta} u = Mg \frac{L}{2} \sin\theta$

$u = \frac{L}{2} \sin^2\theta \quad -(iii)$

$$\tan\theta = \frac{k}{\frac{L}{2}\sin^2\theta}$$

$$\tan\theta \times \sin^2\theta = \frac{2k}{L}$$

$$\frac{\sin^3\theta}{\cos 2\theta} = \frac{2\times 10}{40} = \frac{1}{2}$$

$$2\sin^2\theta = \frac{\cos 2\theta}{\sin\theta}$$

$$\frac{2}{\text{cosec}^2\theta} = \cot\theta$$

∞

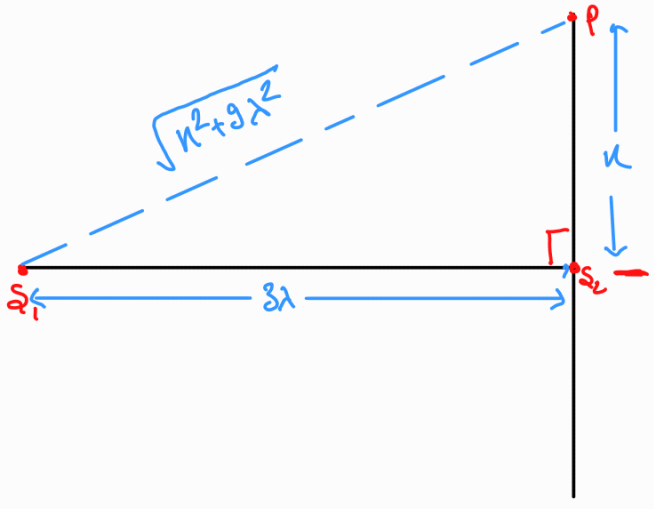
$$\Rightarrow \frac{2}{1+\cot^2\theta} = \cot\theta$$

$$\cot^3\theta + \cot\theta - 2 = 0$$

$$(\cot\theta - 1)(\cot^2\theta + \cot\theta + 2) = 0$$

$$\Rightarrow \cot\theta - 1 = 0 \Rightarrow \theta = 45^\circ$$

(27)



$\Delta n_{max} = 3\lambda \rightarrow$ at S_2

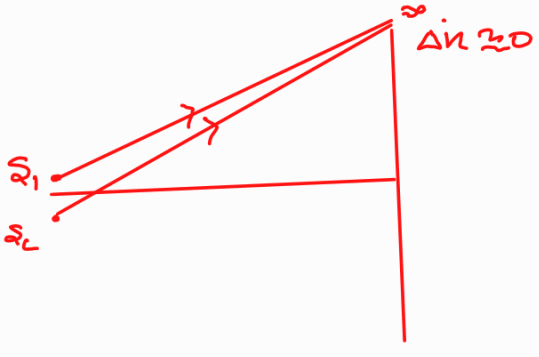
$\Delta n_{min} = 0 \rightarrow$ at ∞

$\Rightarrow$ For minima

$$\Delta n = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}$$

For minimum distance from S_2

$$\boxed{\Delta n = \frac{5\lambda}{2}}$$



$$\Delta n = S_1P - S_2P$$

$$= \sqrt{n^2+9\lambda^2} - n$$

For minimum distance

$$\Delta n = \frac{5\lambda}{2}$$

$$\frac{5\lambda}{2} = \sqrt{n^2+9\lambda^2} - n$$

$$\sqrt{n^2+9\lambda^2} = \frac{5\lambda}{2} + n$$

$$n^2 + 9\lambda^2 = \frac{25\lambda^2}{4} + n^2 + 5\lambda n$$

$$5\lambda n = 9\lambda^2 - \frac{25\lambda^2}{4}$$

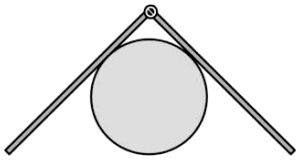
$$5\lambda n = \frac{11\lambda^2}{4} \Rightarrow n = \frac{11\lambda}{20}$$

Now, $\frac{n+5}{20} = \frac{11}{20}$

$$n + 5 = 11$$

$$n = 6$$

cm as shown in the figure. There is no friction between the cylinder and the plates. At what angle (in degree) with vertical will the plates lean in equilibrium?

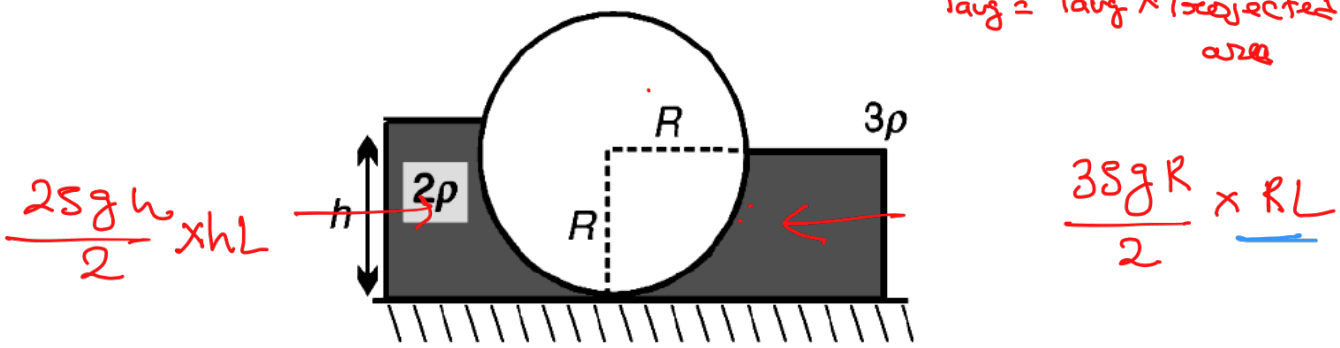


27. Two coherent points sources, S_1 and S_2 vibrating in phase emit light of wavelength λ. The separation between the sources is 3λ. Consider a line passing through S_2 and perpendicular to the line S_1S_2. If the smallest distance from S_2 where a minimum of intensity occurs is $\left(\frac{n+5}{20}\right)\lambda$. find n

28. A man can throw a ball at a speed on the Earth, which can cross a river of width 10m. The man reaches on an imaginary planet whose mean density is twice that of the Earth. Find out the maximum possible radius of the planet(in kilometre) so that if the man throws the ball at the same speed it may escape from the planet. Given radius of earth is $6.4\times10^6 m$

29. In the figure shown, the heavy cylinder $\left(radius\ \sqrt{\frac{8}{3}}m\right)$ kept on a smooth surface separates two liquids of densities 2ρ and 3ρ. The height h(in metre) for the translation equilibrium of the cylinder must be.

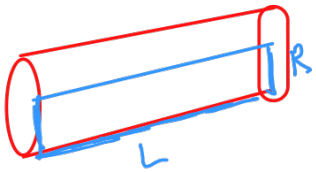
$F_{avg} = P_{avg} \times$ Projected area



$\frac{25gh}{2} \times hL$

$\frac{35gR}{2} \times RL$

30. Two elastic balls of masses m_1 and m_2 are placed on top of each other (with a small gap between them) and then drop onto the ground. The ratio of $m_1\ to\ m_2\ is\ \frac{1}{x}$ for which the upper ball ultimately receives the largest possible fraction of the total energy. Then find the value of x?



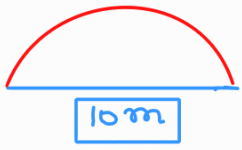
(28) On Earth

$h_{max} = 10m$

$\frac{u^2}{g} = 10m$

$u^2 = 10 \times 10$

$u = 10 m/s$



on the planet

$$v_e = \sqrt{\frac{2GM_p}{R_p}}$$

$$v_e = \sqrt{\frac{2G \times \frac{4}{3}\pi R_p^3 \times \rho_p}{R_p}}$$

$$= \sqrt{\frac{8\pi}{3} G R_p^2 \rho_p}$$

Given, $\rho_p = 2\rho_E$

Let $R_p = kR_E$

$$v_e = \sqrt{\frac{8}{3}\pi G k^2 R_E^2 \times 2\rho_E}$$

$$10 = \sqrt{2}k \times \sqrt{\frac{8}{3}\pi G R_E^2 \rho_E}$$

$$10 = \sqrt{2}\, k \times (11.2 \times 10^3) \text{ m/s}$$

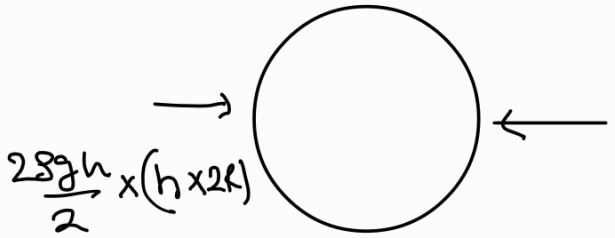
$$k = \frac{10}{\sqrt{2} \times 11.2 \times 10^3}$$

$$R_p = \frac{10}{\sqrt{2} \times 11.2 \times 10^3} \times 6.4 \times 10^6$$

$$= 4040.6 \text{ m}$$

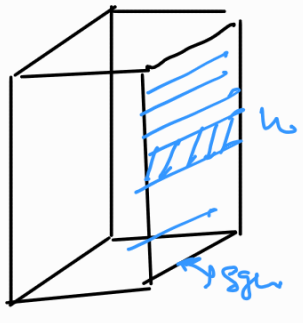
$$\approx 4 Km$$

(29)



$\frac{2\rho gh}{2} \times (h \times 2R)$ ← $\frac{3\rho gR}{2} \times (R \times 2R)$

$$\frac{2\cancel{\rho} gh}{\cancel{2}} \times \cancel{2}Rh = \frac{3\cancel{\rho} gR}{\cancel{2}} \times 2R^2$$



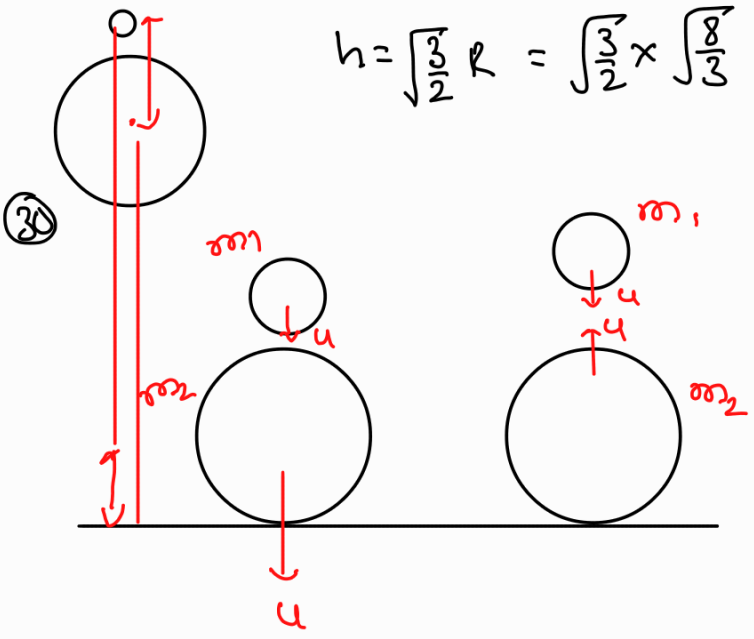
$2h^2 = 3R^2$

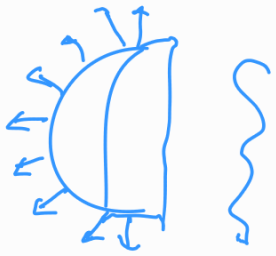
$P_{avg} = \frac{\rho g h}{2}$

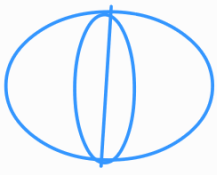
$h = \sqrt{\frac{3}{2}} R = \sqrt{\frac{3}{2}} \times \sqrt{\frac{8}{3}} = \sqrt{4} = 2m$

$F = P_{avg} \times$ Projected area

(30)







In final collision

$$\vec{v}_1 = \frac{\vec{p}_i + e m_2(\vec{u}_2 - \vec{u}_1)}{m_1 + m_2}$$

$$= \frac{(m_2 u - m_1 u) + m_2(u - (-u))}{m_1 + m_2}$$

$$= \frac{m_2 u - m_1 u + 2 m_2 u}{m_1 + m_2}$$

$$= \frac{3m_2 - m_1}{m_1 + m_2} u$$

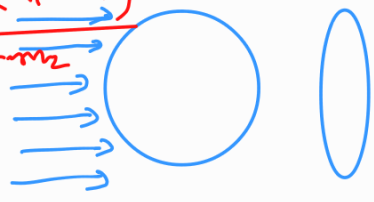
$$v_1 = \frac{3 - \frac{m_1}{m_2}}{\frac{m_1}{m_2} + 1} u$$

Let $n = \frac{m_1}{m_2}$

$$v_1 = \frac{3 - n}{n + 1} u$$

$$E_1 = \frac{1}{2} m_1 v_1^2 = \frac{1}{2} m_1 \left(\frac{3-n}{n+1}\right)^2 u^2$$

$$v_2 = \frac{\vec{p}_i + e m_1(\vec{u}_1 - \vec{u}_2)}{m_1 + m_2}$$



$$E_i = \frac{1}{2}m_1u^2 + \frac{1}{2}m_2u^2 = \frac{1}{2}(m_1+m_2)u^2$$

$$\frac{E_1}{E_i} = \frac{\frac{1}{2}m_1\left(\frac{3-u}{u+1}\right)^2 u^2}{\frac{1}{2}(m_1+m_2)u^2}$$

$$= \frac{(3-u)^2}{(u+1)^2\left(\frac{m_1+m_2}{m_1}\right)}$$

$$= \frac{(3-u)^2}{(1+u)^2\left(1+\frac{1}{u}\right)} = \frac{u(3-u)^2}{(1+u)^3} = \frac{u(9+u^2-6u)}{(1+u)^3} = \frac{u^3-6u^2+9u}{(1+u)^3}$$

For max.

$$\frac{d\left(\frac{E_1}{E_i}\right)}{du} = 0$$

$$\frac{(3u^2-12u+9)(1+u)^3 - (u^3-6u^2+9u)\times 3(1+u)^2(1)}{(1+u)^6} = 0$$

$$3(u^2-4u+3)(1+u) - 3u(u^2-6u+9) = 0$$

$$(u-3)(u-1)(1+u) - u(u-3)^2 = 0$$

$$u^2-1 - u^2+3u = 0$$

$$3u = 1$$

$$u = 1/3$$

(31) $Q_{a-b} = 6P_0V_0$

(a) $W_{ab} = P_0V_0 + \frac{1}{2}P_0V_0 = \frac{3}{2}P_0V_0$

(b) $\Delta U_{ab} = 6P_0V_0 - \frac{3}{2}P_0V_0 = \frac{9}{2}P_0V_0$

(c) $Q_{ab} = nC\Delta T$

$$6P_0V_0 = nC\,\frac{(P_fV_f - P_iV_i)}{nR}$$

$$6P_0V_0R = C\left(4P_0V_0 - P_0V_0\right)$$

$$3P_0V_0C = 6P_0V_0R$$

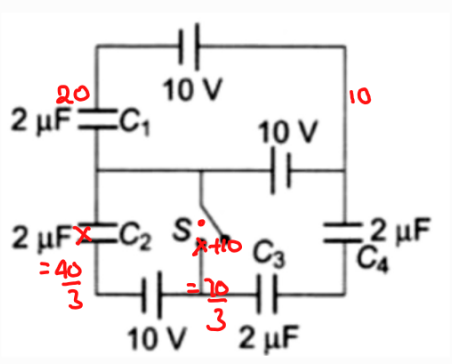
$$C = 2R$$

(d) $\Delta U_{ab} = nC_V\Delta T$

$$\frac{3}{2}P_0V_0 = nC_V\frac{3P_0V_0}{nR}$$

$$C_V = \frac{3R}{2} = 1.5R$$

(32) when S is open



$$(x-20)\times 2 + (x+10-10)\times 1 = 0$$

$$2x - 40 + x = 0$$

$$3x = 40$$

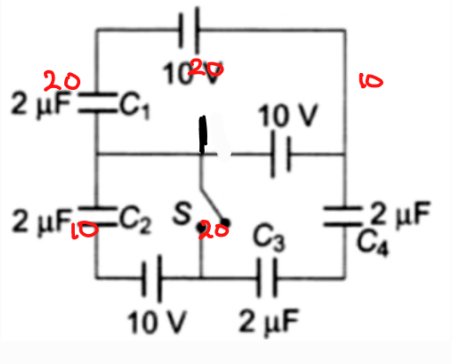
$$x = \frac{40}{3}$$

$$Q_1 = 20\times 2\ \mu C = 40\ \mu C$$

$$Q_2 = \left(20 - \frac{40}{3}\right)\times 2\mu F = \frac{20}{3}\times 2\mu F = \frac{40}{3}\ \mu F$$

$$Q_3 = Q_4 = \left(\frac{70}{3} - 10\right)\times 1\mu F = \frac{40}{3}\ \mu F$$

when S is closed

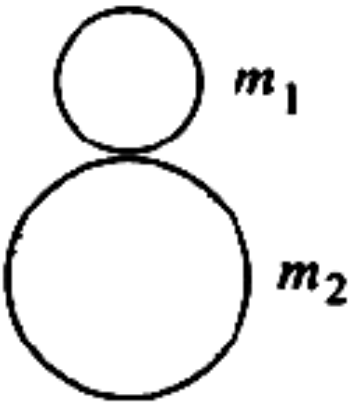


$$Q_1' = 20\times 2\ \mu C = 40\mu C$$

$$Q_2' = (20-10)\times 2\mu C = 20\mu C$$

$$Q_3' = Q_4' = (20-10)\times 1\mu F = 10\mu C$$

$$\Rightarrow Q_1 = Q_1',\ Q_2 < Q_2',\ Q_3 > Q_3',\ Q_4 > Q_4'$$



SECTION – IV
(MATCHING TYPE)

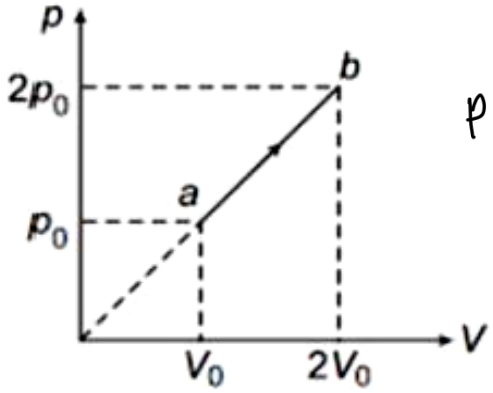
This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists :**List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

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31. Heat taken by a gas in process $a \to b$ is $6P_0V_0$ .Match the following columns.



$p \propto \frac{1}{r}$ = Process

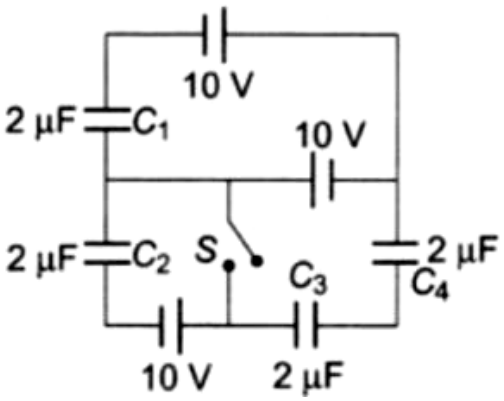
Column I	Column II
(a) W_{ab}	(p) $2P_0V_0$
(b) ΔU_{ab}	(q) $4P_0V_0$
(c) Molar heat capacity in given process	(r) $2R$
(d) C_v of gas	(s) None of these

A) $a \to$s ,b$\to s$,$c \to r$, $d \to s$ **B)** $a \to$s ,b$\to q$,$c \to r$, $d \to s$

C) $a \to$s ,b$\to p$,$c \to s$, $d \to r$ **D)** $a \to$q ,b$\to s$,$c \to r$, $d \to s$.

32. Before connecting the circuit shown in figure, all capacitors were uncharged. The circuit now is in steady state.

Now switch S is closed. During the time, the study state is reached again, match the following:

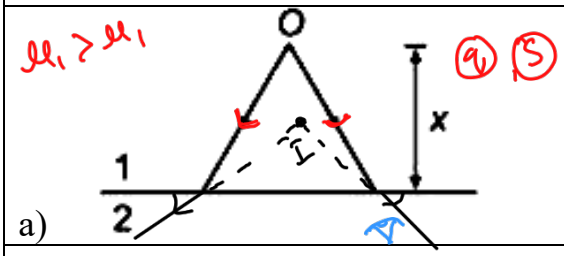
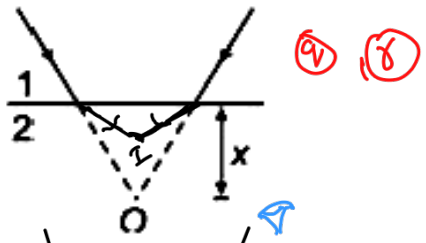
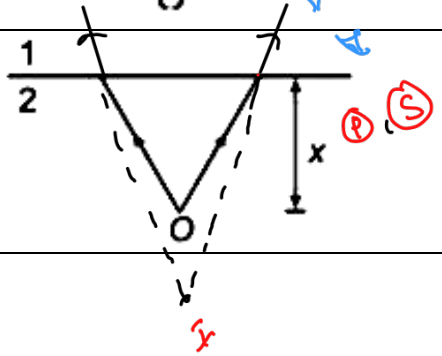


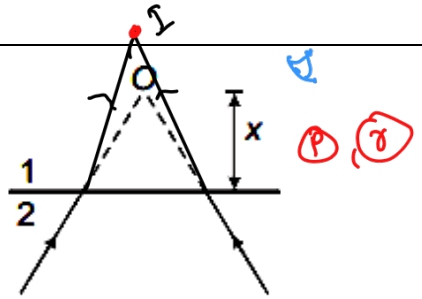
Column I	Column II
a) charge on c_1	p) increases
b)charge on c_2	q) decreases
c)charge on c_3	r) remains same
d)charge on c_4	s) becomes zero

A) $a \rightarrow$r , b$\rightarrow s$,$c \rightarrow r$,$d \rightarrow q$ **B)** $a \rightarrow s$,$b \rightarrow p$,$c \rightarrow r$,$d \rightarrow s$

C) $a \rightarrow$r ,b$\rightarrow p$,$c \rightarrow q$,$d \rightarrow q$ **D)** $a \rightarrow q$,$b \rightarrow s$,$c \rightarrow r$,$d \rightarrow s$

33. Match the following two columns corresponding to single refraction from plane surface . In all cases shown in column I, $\mu_1 > \mu_2$

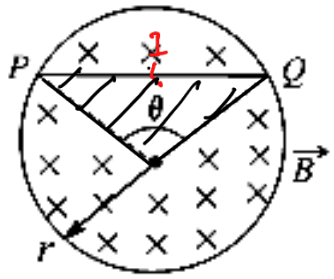
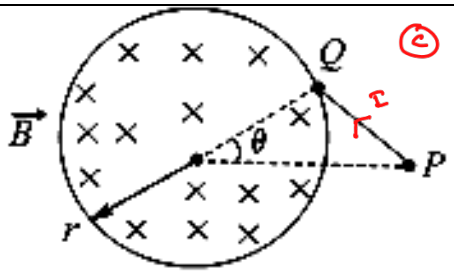
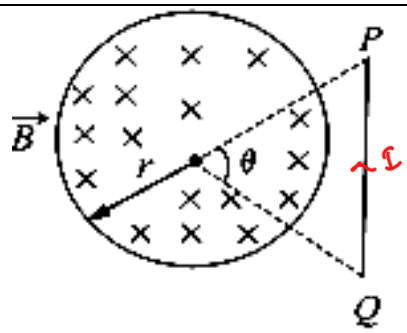
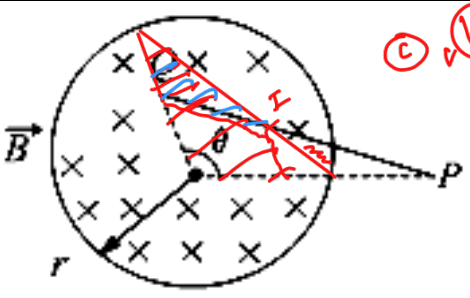
Column I	Column II
a) 	p) Image distance greater than x from plane surface
b) 	q) Image distance less than x from plane surface
c) 	r) Real image

d) 	s) Virtual image

A) $a\rightarrow q,s;\ b\rightarrow q,r;\ c\rightarrow p,s;\ d\rightarrow p,r$ **B)** $a\rightarrow q,s;\ b\rightarrow q,r\ c\rightarrow p,r;\ d\rightarrow p,s$

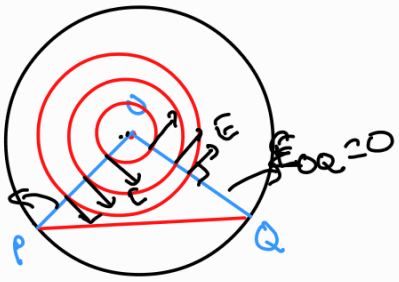
C) $a\rightarrow q,r;\ b\rightarrow q,s;\ c\rightarrow p,s;\ d\rightarrow p,s$ **D)** $a\rightarrow p,s;\ b\rightarrow q,r\ ;\ c\rightarrow p,s;\ d\rightarrow q,r$

34. Column I shows the cylindrical region of radius r where a downward magnetic field $\vec{B}$ exists, where $\vec{B}$ is increasing at the rate of $\frac{d\,B}{dt}$. A rod PQ is placed in different citation as shown. Match the column I with the correct statement in column II regarding the induced emf in the rod.

Column I	Column II
i. 	a. Induced emf in rod PQ is $\boxed{\frac{1}{2}r^2\theta\frac{dB}{dt}}$.
ii. 	b. Induced emf in rod PQ is less than $\frac{1}{2}r^2\theta\frac{dB}{dt}$.
iii. 	c. End P is positive with respect to point Q.
iv. 	d. End Q is positive with respect to point P.

A) $i\rightarrow b,c;ii\rightarrow a,c;\ iii\rightarrow a,d;\ iv\rightarrow b,d$ **B)** $i\rightarrow b,c;\ ii\rightarrow a,d;\ iii\rightarrow a,c;iv\rightarrow b,d$

C) $i\rightarrow b,c;\ ii\rightarrow a,c;\ iii\rightarrow a,d;\ iv\rightarrow b,c$ **D)** $i\rightarrow b,d;\ ii\rightarrow a,c;\ iii\rightarrow a,d;\ iv\rightarrow b,c$



$\mathcal{E}_{OP} = 0$

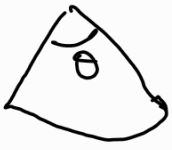
$\mathcal{E}_{OQ} = 0$

$\phi = BA$

$\mathcal{E} = \frac{d\phi}{dt} = A\frac{dB}{dt}$

$\mathcal{E}_{OP} + \mathcal{E}_{PQ} + \mathcal{E}_{QO} = A\frac{dB}{dt}$ ($\mathcal{E}_{OP} = 0$, $\mathcal{E}_{QO} = 0$)

$\mathcal{E}_{PQ} = A\frac{dB}{dt}$



$\frac{\theta}{2\pi} \cdot \pi r^2$

$\frac{\theta}{2\pi} \cdot \pi r^2 = \frac{1}{2}\theta r^2$

$\frac{1}{2} r^2 \theta \frac{dB}{dt}$

(33) For (a) & (b) when ray is going from 1 to 2

$$d_{app} = \frac{\mu_2}{\mu_1} d$$

& $\mu_1 > \mu_2$

$\Rightarrow d_{app} < d$

For (c) & (d) when ray is going from 2 to 1

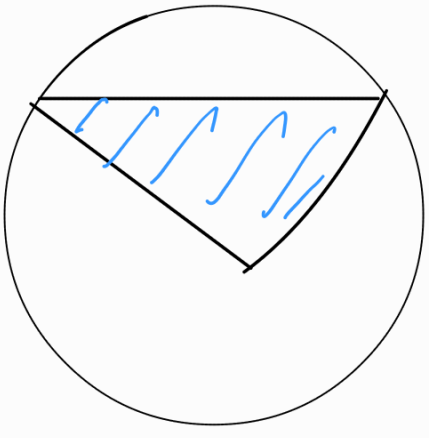
$$d_{app} = \frac{\mu_1 . d}{\mu_2}$$

& $\mu_1 > \mu_2$

$d_{app} > d$

Extend the rays, if they cut in real then image is real & if we need to extend them then image is virtual

(34)



$$\mathcal{E} = A \frac{dB}{dt}$$

$$A < \frac{1}{2} \theta r^2$$